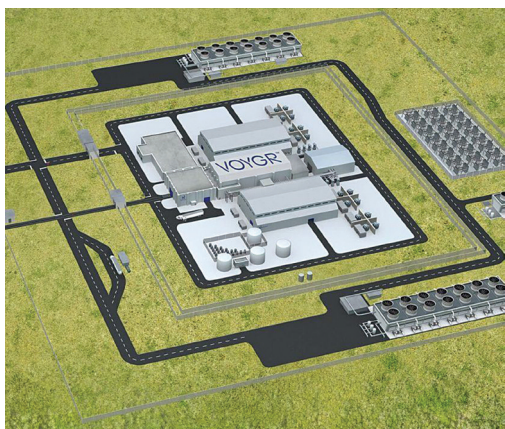




Graphic depiction of the future VOYGR power plant
(Source: Nuscalepower)

Assured policy support, regulatory reform, and risk-sharing mechanisms are essential to attract private and foreign investment.

Integrating SMRs into existing energy infrastructure necessitates grid upgrades and site modifications, requiring substantial investment. Repurposing former coal plant sites for SMRs requires modifications tailored to the specific needs of SMRs.

Unlike conventional reactors that use established fuel cycles, advanced SMR designs require new fuel types such as high-assay low-enriched uranium (HALEU) and Tristructural-Isotropic (TRISO) fuels. To ensure their timely commercial availability, dedicated supply chains and regulatory frameworks must be developed in parallel.

Serious public scepticism persists due to lingering concerns around nuclear safety and radioactive waste disposal. Addressing these concerns requires transparent policymaking, regular public engagement, and educational initiatives that communicate the safety and environmental benefits of SMRs.

Overcoming these challenges demands a concerted and sustained effort involving governments, industry stakeholders, financial institutions, and the research

community. Adapting regulatory frameworks to suit SMR characteristics, developing innovative financial models, establishing secure and resilient fuel supply chains, and fostering informed public engagement are key priorities.

Together, these efforts will help position SMRs as a reliable, scalable, and sustainable energy source.

The way ahead and the future

With global demand for electricity expected to double by 2050, the urgency for a sustainable energy transition has become imperative. While renewable sources such as wind and solar power are expanding rapidly, their inherent intermittency underscores the need for consistent, round-the-clock baseload power.

SMRs represent a transformative step in nuclear energy and can serve as standalone systems or complement renewable energy sources in hybrid energy models. Their phased construction, shorter build time, and lower capital and operational costs make them suitable for diverse applications such as hydrogen production, district heating, desalination, and industrial process heat.

Globally, momentum is building around SMR development. Several designs have received regulatory approval or certification. Russia has deployed the world's first floating SMR power plant, Akademik Lomonosov. Construction is underway in China for the ACP100 reactor and in Argentina for the CAREM-25 SMR project. The United States has made progress with demonstration projects such as NuScale's VOYGR SMRs.

Canada is also supporting

multiple SMR projects. Collectively, over 300 SMRs are expected to be deployed worldwide by 2040, backed by estimated investments exceeding \$100 billion. This is projected to create employment for approximately one million skilled professionals.

Emerging advanced SMR technologies, such as high-temperature reactors and mobile reactors, offer enhanced capabilities for specific high-demand applications. High-temperature reactors will enable industrial processes that have previously been difficult to achieve. Mobile reactors will deliver power to remote locations, military bases, and disaster zones—where it is needed most. For example, mobile reactors may provide on-site thermal energy for oil recovery processes, moving alongside extraction operations.

Transparent policymaking, streamlined regulations, and targeted educational outreach are required to overcome civil society's scepticism. Realising the full potential of SMRs and building public trust will require close collaboration among governments, industry leaders, research institutions, and civil society.

However, challenges remain. Existing nuclear licensing and regulatory frameworks need to be adapted to meet the specific requirements of SMRs. To ensure timely and cost-effective deployment, it is necessary to modernise infrastructure, strengthen the nuclear supply chain, and streamline processes for refuelling, waste management, and maintenance.

With the right policy environment, rapid technological progress, and consistent societal engagement, SMRs can play a pivotal role in delivering reliable, sustainable, and equitable energy for decades to come. **EFY**